Million-Parameter Python Code Generation on Microcontrollers: Signature Anchoring and Compiler-Gated Repair

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***Abstract*—Can a code model small enough for microcontroller flash write useful embedded utility functions? We train decoder-only Transformers with 0.53 M and 1.05 M parameters on a leak-audited benchmark of IoT-oriented Python tasks and run them with a portable INT8 engine on a PC and in ESP32-S3 firmware. On unseen wordings of known tasks, the 0.53 M model, which needs 588,504 bytes of flash, passes the unit tests on 82.4 % of tasks with *signature anchoring* and 85.7 % with compiler-gated repair (mean of 3 seeds). Repair mainly fixes syntax, is no better than resampling at equal cost, and does not need the error message. On unseen task families accuracy falls to at most 2.4 %: such models are verifiable specialists for a closed catalogue of device functions, not general programmers.**

***Index Terms*—TinyML, code generation, small language models, program repair, microcontrollers.**

I. Introduction

Code-generating language models [1], [2] have billions of parameters, while microcontrollers have megabytes of memory and are often offline [3]–[5]. Even "small" models such as MobileLLM [6] have hundreds of millions of parameters, but TinyStories [7] showed that a few million parameters suffice for fluent text in a narrow domain. We ask what a code model that fits in microcontroller flash can do, and which inexpensive decoding techniques make it more reliable. A compiler detects syntax errors for free, and feeding its diagnostics back to a model is established for large models [8]–[10]; yet at equal sampling budget, self-repair is often no better than drawing more samples [11]. We test this at a scale about 10^4 times smaller, where every extra generation costs time and energy on the device.

Contributions. (i) A benchmark of 48 training and 16 held-out embedded task families with executable tests and an automated leakage audit. (ii) Transformers of 0.53 and 1.05 M parameters with an exact flash and RAM budget, deployed through a dependency-free INT8 C engine and ESP-IDF firmware. (iii) An equal-budget comparison of signature anchoring, compiler-gated repair with and without the error message, and resampling, over three seeds. (iv) Negative results reported as such: no generalisation to unseen families or to MBPP [12].

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Code, data, models and firmware: https://github.com/rohithyarramala/microcoder (archived: https://doi.org/10.5281/zenodo.24187363). An extended preprint is available at https://doi.org/10.5281/zenodo.24189966.

II. Related Work

Small and on-device models. TinyML deploys networks on microcontrollers through quantisation [13], specialised runtimes [4] and joint architecture–runtime design [5], with benchmarks for vision and audio [14]. For language, narrow high-quality data lets small models punch above their size [7], [15], but compact models such as TinyLlama [16] and MobileLLM [6] target phones, and pretrained code models such as CodeT5 [17] still have tens of millions of parameters. We are not aware of code-generation results for models that run from microcontroller flash.

Repair from compiler feedback. DeepFix [8] and DrRepair [9] train models to fix compiler errors, Break-It-Fix-It [18] learns from a model's own mistakes, and Self-Debug [10], Self-Refine [19] and Reflexion [20] prompt large models with feedback; see [21] for program repair in general. Olausson et al. [11] showed that repair must be compared with resampling at equal cost. We follow that protocol, and we also tested the Break-It-Fix-It idea: on training prompts the model produced only 1 syntax error in 15,328 samples, so there is nothing to learn from without leaking test wordings.

III. Benchmark, Model and Decoding

Tasks and splits. A *family* is a parameterised task with four natural-language phrasings, a reference solution and a test-input generator; tasks sample function names, argument names and constants (e.g. ADC resolution, filter coefficient, bit index). The 64 families cover sensor/IoT utilities (scaling, filtering, debouncing, checksums, byte packing, bit tests), math, list, string and recursion tasks. With a fixed seed, 16 families are held out from training. The *in-family* test split (384 tasks) uses phrasing 4 of the training families, which is never trained on; the *out-of-family* split (240 tasks) uses the held-out families. Design choices were made on a separate development split (phrasing 3). A script verifies zero exact or 8-gram instruction overlap between training and the in-family test split and zero held-out families in training. A task passes if the program returns the expected value on all five test inputs in a sandbox with a 2 s time limit.

Model. A decoder-only Transformer [22] with RMSNorm [23], rotary embeddings [24], GELU MLP, no biases and tied embeddings [25], and a byte-level BPE tokenizer [26] with 2,048 entries. Model S has 3 layers of width 96 (529,056 parameters) and model M has 4 layers of width 128 (1,049,728 parameters). Training uses 7,664 instruction–code pairs and

TABLE I
BUDGET PER GENERATED TOKEN FOR MODEL S (0.53 M) AND M (1.05 M). HOST: ONE X86 CORE, SAME C ENGINE. ESP32-S3 LATENCY IS AN ANALYTICAL ESTIMATE, NOT A MEASUREMENT.

	S	M
Flash: INT8 model + tokenizer (B)	588,504	1,118,680
RAM: state incl. KV cache (B)	626,688	1,094,656
MACs/token, position 256	675,840	1,310,720
Host decode (ms/token)	0.373	0.708
ESP32-S3 decode, estimate (ms/token)	10–24	–

39,434 synthetic repair triples (instruction, corrupted program, CPython error, correct program), with the loss on answer tokens only, in 18–33 min per model on a CPU, without a GPU.

Decoding (at most four generations per task). *Signature anchoring* (SA) extracts the first `name(args)` from the request and forces the answer to begin with `def name(args):`. *Recursive Error-Feedback* (REF) re-prompts the model with the request, the faulty draft and the error message while a gate rejects the program; the gate is `compile()`, optionally followed by a 118-line static check for names that are read but never bound, which also runs in MicroPython. We compare the full CPython message, a line-only message (roughly what MicroPython reports) and no message, and the equal-budget baseline of [11]: resample until the gate passes.

IV. DEPLOYMENT ON THE ESP32-S3

Weights are quantised per output row to INT8 with FP32 scales; activations and the KV cache stay FP32 for the ESP32-S3's single-precision FPU. The C99 engine reads weights in place from flash, does not allocate in the decode loop and includes the tokenizer; it reproduces the reference tokenizer on all 18,524 benchmark and corpus strings, and its logits match PyTorch with the same quantised weights to within 10^{-4}. Running the full benchmark through the INT8 engine changes the SA pass rate of model M from 70.8 to 70.8 % and of model S from 81.5 to 81.5 %. Built with ESP-IDF v5.4.2, the firmware image with model S and all 624 test prompts is 870,176 bytes; static internal-RAM use is 55,871 of 341,760 bytes, and the KV cache for the 256-token context (589,824 bytes) requires PSRAM, which most ESP32-S3 modules have. The unmodified `app_main()`, run on a PC under the same memory budget, passes 51.9 % of the 624 prompts with SA and greedy decoding, matching PyTorch. A MicroPython module runs the REF loop on the device with MicroPython's own `compile()` as the gate.

Table I gives the budget. We have not yet measured latency on ESP32-S3 hardware; the estimate assumes 4 cycles per INT8 multiply–accumulate at 240 MHz plus streaming all weights from external flash at 40 MB/s per token, and a board run (`esp32/run_lab.sh board` in the repository) replaces it with measured latency and on-device accuracy.

V. RESULTS

TABLE II
PASS RATE (%), MEAN ± S.D. OVER 3 SEEDS. ID: IN-FAMILY (384 TASKS); OOD: OUT-OF-FAMILY (240 TASKS), MODEL M. "GENS": MEAN GENERATIONS USED (BUDGET 4), MODEL M.

Strategy	S, ID	M, ID	M, OOD	Gens
Greedy	70.9±1.9	67.9±12.5	0.7	1.00
SA	82.4±1.3	77.5±7.1	1.2	1.00
SA + resample	–	77.9±7.1	1.4	1.12
SA + REF, full msg.	84.0±1.2	78.3±6.8	1.2	1.10
SA + REF, line only	–	78.4±6.8	1.2	1.10
SA + REF, no msg.	–	78.6±6.7	1.4	1.10
SA + REF, two gates	85.7±2.0	79.8±6.0	1.2	1.26

Signature anchoring is the largest gain (Table II). For model M it raises the in-family pass rate from 67.9 to 77.5 % at no cost: without it, the model often writes a correct body under a slightly different header, which fails every test.

Repair fixes syntax, not behaviour, and the message does not matter. REF raises syntactic validity from 95.2 to 97.7 % in-family and from 69.2 to 86.0 % out-of-family, but the pass rate barely moves. With the full message, a line-only message and no message, SA+REF reaches 78.3, 78.4 and 78.6 %; resampling reaches 77.9 %. All differences are far smaller than the seed spread, which matches the large-model finding of [11]: the repair prompt helps by giving a second conditioned attempt, not because the model reads the diagnostic. For a device, this means the cheaper line-only message of MicroPython loses nothing.

A second free gate helps a little. Of the 108 in-family failures of SA+REF (seed 0), 28 are `NameErrors` that `compile()` accepts. Gating on `compile()` plus the name check gives the best result, 79.8 % for model M with 1.26 generations on average; the check has no false positives on all 5,581 reference solutions. The gain is consistent across seeds but smaller than their spread.

Half a megabyte is enough. Model S is at least as accurate as model M (82.4 ± 1.3 % with SA and 85.7 % with SA+REF and two gates) and is more stable across seeds; a 3.05 M model reaches 69.0 % with SA (one seed). For a closed task catalogue, model size is not the bottleneck. Training without the repair triples lowers even greedy accuracy (45.6 vs. 59.6 %, one seed), so repair data also acts as extra supervision. Augmenting instructions with random identifiers *lowered* development-split accuracy, because descriptive names are a strong cue to the intended task.

No generalisation beyond the catalogue. On held-out families every strategy passes 0.7–2.4 % of tasks (both models), although REF makes most programs compile. On the 500 MBPP test problems it writes valid Python for 47.8 % of them but passes 0. The model recalls and adapts known tasks; it does not compose new algorithms.

VI. DISCUSSION AND CONCLUSION

A model of about half a megabyte can serve as an offline, verifiable generator for a closed catalogue of device functions, such as unit conversions, filters, thresholds, checksums and bit operations, adapted by a technician in the field. At this scale the decoding procedure matters more than error feedback: signature anchoring plus one gated retry captures nearly all of the

available accuracy. *Limitations:* the benchmark is synthetic, and 355 of the 384 in-family reference solutions also occur in training, so the in-family split tests recognition of known tasks from unseen wording; the seed spread is large; the gates check form, not function; and on-device latency is estimated, not yet measured. Code, data, models, firmware and the scripts that generate every number in this letter are public.

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